

# ANDREW CHU

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## EDUCATION

### University of Waterloo

*Bachelor of Computer Science (Honours)*

Coursework: Compilers, Object-Oriented Programming, Functional Programming, Data Structures & Algorithms

Apr. 2028

Waterloo, ON

## TECHNICAL SKILLS

**Languages:** TypeScript, Python, C/C++, Java, JavaScript, Go, Ruby, CSS, HTML, Bash, Kotlin

**Frameworks:** Next, React, Express, Flask, Redis, Jest, SvelteKit, Electron, LangChain, MongoDB, Supabase

**Developer Tools:** Azure, PyTorch, Pandas, Linux, Git, Figma, Docker, GCP, Nginx, Node, Prometheus, Grafana

## EXPERIENCE

### Software Engineer Intern

May 2026 - Aug. 2026

*Stripe*

*Toronto, ON*

- Developed suite of synthetic tests for critical payment and device attestation flows, preventing **5+** historic incidents for critical microservice processing **\$100M+**/month while balancing testability and security integrity
- Drove proposal, implementation, and deliverables end-to-end for backend microservices and Kotlin SDK features
- Shipped API and SDK integration for alerting attestation failures, enabling self-triage of **30K+ errors/month**
- Implemented **10** company-wide bug fixes, including fixing a broken coding MCP tool with **100K+ calls/day**

### Software Engineer Intern

Sep. 2025 - Dec. 2025

*Alljoined*

*San Francisco, CA*

- Developed unified infrastructure for processing EEG data, saving research and ML teams **30+ hours/week**
- Migrated **2+ million** EEG trials with **0%** downtime from live database to custom lakehouse using dual-writes
- Automated internal data labeling system using LLMs with **98%** accuracy, saving **300+** hours of manual work
- Productionized EEG decoding research model, reducing fine-tuning time by **87%** and improving accuracy by **12%**

### Software Engineer Intern

May 2025 - Aug. 2025

*CGI*

*Markham, ON*

- Developed multimodal RAG system to retrieve information from **100+** internal production guides, accelerating team incident response times by **17%**
- Increased retrieval and response accuracy by **38%** through PEFT fine-tuning a Llama3 model with QLoRA
- Automated health checks, server restarts, and email notifications, by engineering unified dashboard for remotely executing jobs on **200+** applications and servers with Ansible

### Data Engineer Intern

Aug. 2024 - Sep. 2024

*Alljoined*

*Toronto, ON*

- Debugged critical timing issue in data preprocessing by implementing data observability system in Python
- Reduced latency in data collection by **95%** by refactoring WebSocket I/O to prevent deadlocking
- Designed signal processing pipeline for denoising and artifact correction using NumPy, Pandas, and scikit-learn

### Full-Stack Developer

Sep. 2022 - Jan. 2023

*Digitera Interactive*

*Ottawa, ON*

- Developed "Skule News" project, a mobile school news app to streamline student communication written in Flutter
- Integrated Firebase for authentication, real-time updates, and cloud file storage for secure backend support
- Reduced frequency of in-app crashes by **20%** by resolving **6+** bugs resulting from incorrect Firebase queries

## PROJECTS

### Moonlight - Prediction Market Trading Bot | *Grafana, Prometheus, Docker, AWS, FastAPI, MongoDB, Python, Polymarket*

- Developed microservice-based, idempotent trading bot, executing trades 24/7 with **80ms P95 latency**
- Productionized deployment on AWS EC2 using Docker and Prometheus/Grafana monitoring for visibility

### Replate - Sustainable Delivery App (GenAI Genesis Winner) | *Next, Express, MongoDB, Flask, Cohere, Gemini, Twilio*

- Achieved **AI Eco-Mobility Award** at GenAI Genesis, Canada's largest AI hackathon, among **600+** competitors
- Engineered a meal recommendation system feeding *Too Good To Go's* API into an evaluator-optimizer workflow
- Utilized Express and Flask backend microservices to process orders and serve live delivery updates via Twilio SMS